Fiscal Year:	FY 2008	**Task Last Updated:**	FY 03/03/2008
PI Name:	Lathan, Corinna E. Ph.D.		
Project Title:	Mental Representation of Spatial Cues (3D-Space)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20910	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	ILSRA 2001
Start Date:	05/02/2005	**End Date:**	09/30/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
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Flight Program:	ISS		
Flight Assignment:	ISS Co-Investigator to ESA managed flight study that is in flight development phase.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clement, Gilles (CNRS (Centre national de la recherche scientifique))		
Grant/Contract No.:	NNJ05HD92A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The purpose of this study is to investigate the effects of acute and long-term effects of exposure to microgravity on the mental representation of spatial cues by astronauts during and after space flight. Previous space studies carried out by our group have shown that cognitive functions such as mental rotation, symmetry detection, and the internal representation of the vertical axis of objects were found to be impaired during and after space flight. We propose to repeat the preliminary observations on a larger number of human subjects during short-and-long-term space missions, by using simple experiments requiring a laptop computer and a digitizing table. The analysis of the spatial and dynamic components of writing and drawing tests of 2-D and 3-D objects performed throughout a space mission should allow the researchers to confirm that the internal spatial representation of the vertical dimension, which is determined on Earth by the gravitational reference system, is altered during space flight. If crew time and equipment availability permit, additional psychophysics measurements will be made using a portable virtual reality system displaying 3-D objects with one dimension altered. Tests of the perception of absolute distances will also be performed using the same equipment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this study could have important consequences for human performance during space flight. For example, if an astronaut imagines a space station shorter than it is in reality, the expected duration of a trip within this station will be shorter than what it actually takes, thus leading to delays and stress. There may also be consequences for a space habitat design if squared volumes do not look square to astronauts. Virtual reality training may be a way to train the astronauts to compensate for such altered spatial representation.
Task Progress:	During the current reporting period, software to support the assessment of mental representation of spatial cues during spaceflight was completed and tested for use in ground-based and spaceflight experiments. The final versions of both the training and experimentation software were transitioned to ESA, and a complete software user guide with instructions for installation and use of the 3D-SPACE software was delivered. The associated hardware for use in ground-based and spaceflight experimentation was qualified for spaceflight by CNES. Additionally, experimental design and administrative tasks were completed during this period of performance, including coordination and scheduling of training and Baseline Data Collection (BDC) for the Increment 17 crew. Software development efforts for the current reporting period included modifications to the individual 2D and 3D task components, as well as to the data collection and reporting architecture of the flight software. The Graphical User Interface (GUI) was also modified to support ease of use with minimal training. The initial 2D and 3D task components were developed for parabolic flight testing during the previous reporting period and were modified based on feedback and observations during testing. During initial testing, data was collected and saved locally to a text file for post hoc analysis. For the flight software, a database was created to save and pre-process data in near real-time, including automated percent error calculations. Training software consisting of a similar series of tasks to those presented in the experimentation software was designed and implemented as well to ensure the experimentation stimuli would be novel. Additionally, a significant effort was required to implement a software solution to prevent the right and left eye images on the 3D goggles from becoming swapped. A series of tests were completed to validate the software. Additionally, qualification of the experiment hardware was completed by CNES. Administrative tasks conducted during the current reporting period included weekly team meetings and teleconferences with the ESA PI, progress documentation and reporting, travel to and from meetings and experiments, and coordination of the training and Baseline Data Collection (BDC) for the Increment 17 crew.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Clément G, Lathan C, Lockerd A. "Perception of depth in microgravity during parabolic flight." Acta Astronautica. 2008 Oct-Nov. 63(7-10):828-32. https://doi.org/10.1016/j.actaastro.2008.01.002 , Oct-2008